

Absolute value inequalities

MathE

19th of April of 2023

The **absolute value** of a number $x \in \mathbb{R}$ is defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Geometrically, you can think of in the following way:

- Given $x \in \mathbb{R}$, you can represent x on the real line.

- If $x = 0$, it coincides with the origin of the real line.



- If $x > 0$, it is to the right of the origin (we are considering the numbers are increasing from left to right).

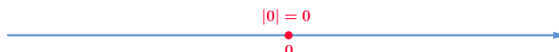


- If $x < 0$, it is to the left of the origin.

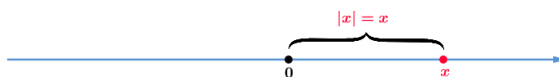


- The **absolute value** of x is the distance from x to the origin.

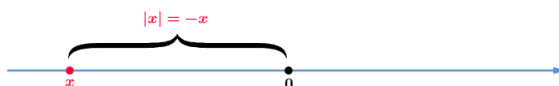
- If $x = 0$, the distance from 0 to 0 is 0, so $|0| = 0$.



- If $x > 0$, then the distance from x to 0 (the origin) is x .



- If $x < 0$, then the distance from x to 0 (the origin) is $-x$ (note that $x < 0$ implies that $-x > 0$).



Exercise 1. Solve the inequality $|x| < 5$.

As we saw above, this means that **the distance from x to the origin** is less than 5.



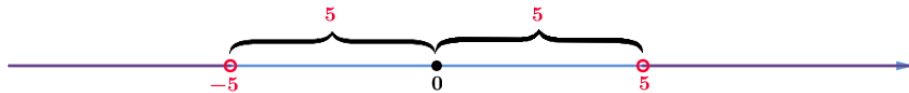
In other words, $|x| < 5$ means that we have to have simultaneously $x < 5$ **and** $x > -5$:

$$|x| < 5 \iff x < 5 \wedge x > -5 \iff -5 < x < 5$$

Answer: $-5 < x < 5$ or $x \in (-5, 5)$

Exercise 2. Solve the inequality $|x| > 5$.

As we saw above, this means that **the distance from x to the origin** is greater than 5.



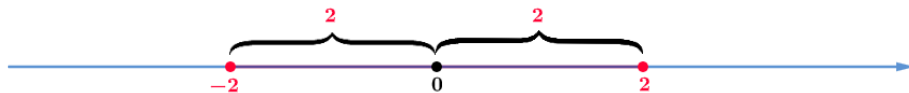
In other words, $|x| > 5$ means that we have to have $x < -5$ **or** $x > 5$:

$$|x| > 5 \iff x > 5 \wedge x < -5$$

Answer: $x > 5 \wedge x < -5$ or $x \in (-\infty, -5) \cup (5, +\infty)$

Exercise 3. Solve the inequality $|3x - 7| \leq 2$.

As we saw above, this means that **the distance from $3x - 7$ to the origin** is less than or equal to 2.



In other words, $|3x - 7| \leq 2$ means that we have to have simultaneously $3x - 7 \leq 2$ **and** $3x - 7 \geq -2$:

$$\begin{aligned} |3x - 7| \leq 2 &\iff \\ 3x - 7 \leq 2 \wedge 3x - 7 \geq -2 &\iff \\ 3x - 7 + 7 \leq 2 + 7 \wedge 3x - 7 + 7 \geq -2 + 7 &\iff \\ 3x \leq 9 \wedge 3x \geq 5 &\iff \end{aligned}$$

$$\begin{aligned} \cancel{3}x \leq \frac{9}{\cancel{3}} \wedge \frac{\cancel{3}x}{\cancel{3}} &\geq \frac{5}{\cancel{3}} \iff \\ x \leq 3 \wedge x &\geq \frac{5}{3} \end{aligned}$$



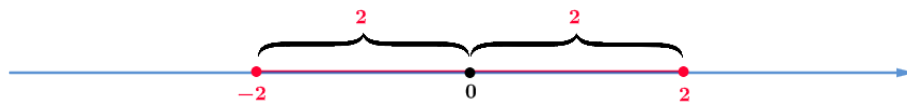
Answer: $x \leq 3 \wedge x \geq \frac{5}{3}$ or $x \in [\frac{5}{3}, 3]$

Exercise 4. Solve the inequality $5|4 - 3x| - 10 \leq 0$.

In this type of inequalities, the first step is to isolate the modulus on one of the sides of the inequality:

$$\begin{aligned} 5|4 - 3x| - 10 &\leq 0 \iff \\ 5|4 - 3x| &\leq 10 \iff \\ |4 - 3x| &\leq 2 \end{aligned}$$

As we saw above, this means that **the distance from $4 - 3x$ to the origin** is less than or equal to 2.



In other words, $|4 - 3x| \leq 2$ means that we have to have simultaneously $4 - 3x \leq 2$ **and** $4 - 3x \geq -2$:

$$\begin{aligned} |4 - 3x| &\leq 2 \iff \\ 4 - 3x &\leq 2 \wedge 4 - 3x \geq -2 \iff \\ \cancel{4} - \cancel{4} - 3x &\leq \cancel{-4} + 2 \wedge \cancel{4} - \cancel{4} - 3x \geq \cancel{-4} - 2 \iff \\ -3x &\leq -2 \wedge -3x \geq -6 \stackrel{(1)}{\iff} \\ \frac{-3x}{-3} &\geq \frac{-2}{-3} \wedge \frac{-3x}{-3} \leq \frac{-6}{-3} \iff \\ x &\geq \frac{2}{3} \wedge x \leq 2 \end{aligned}$$



(1) Our goal is to isolate x on one of the sides of the inequalities.

In order to do so, in this step we must divide both terms of the inequalities by -3 , because this allow us to “get rid” of -3 .

However, when we divide the terms of an inequality by a **negative number**,

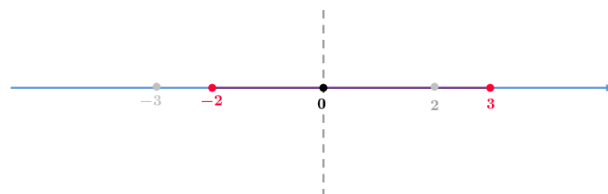
- $\leq$ changes to $\geq$
- $\geq$ changes to $\leq$
- $<$ changes to $>$
- $>$ changes to $<$

because the minus sign implies a **reflection** on the origin. For example, if we have

$$-3 \leq x \leq 2$$



and if we divide all the terms by -1 , we get



so $-2 \leq -x \leq 3$.

Answer: $x \leq 2 \wedge x \geq \frac{2}{3}$ or $x \in [\frac{2}{3}, 2]$

Exercise 5. Solve the inequality $\frac{5}{4} - \left| \frac{1}{2} - \frac{1}{4}x \right| < \frac{3}{8}$.

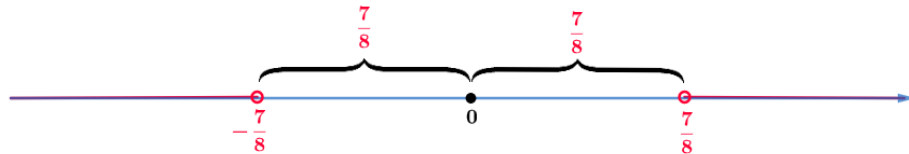
In this type of inequalities, the first step is to isolate the modulus on one of the sides of the inequality:

$$\begin{aligned} \frac{5}{4} - \left| \frac{1}{2} - \frac{1}{4}x \right| &< \frac{3}{8} \iff \\ \cancel{\frac{5}{4}} + \frac{5}{4} - \left| \frac{1}{2} - \frac{1}{4}x \right| &< \underbrace{\cancel{\frac{5}{4}}}_{(\times 2)} + \frac{3}{8} \iff \\ -\left| \frac{1}{2} - \frac{1}{4}x \right| &< -\frac{10}{8} + \frac{3}{8} \iff \\ -\left| \frac{1}{2} - \frac{1}{4}x \right| &< -\frac{7}{8} \iff \end{aligned}$$

$$\frac{-\left|\frac{1}{2} - \frac{1}{4}x\right|}{-1} > \frac{-\frac{7}{8}}{-1} \iff$$

$$\left|\frac{1}{2} - \frac{1}{4}x\right| > \frac{7}{8}$$

As we saw above, this means that **the distance from $\left|\frac{1}{2} - \frac{1}{4}x\right|$ to the origin** is greater than $\frac{7}{8}$.



In other words, $\left|\frac{1}{2} - \frac{1}{4}x\right| > \frac{7}{8}$ means that we have to have

$$\frac{1}{2} - \frac{1}{4}x < -\frac{7}{8} \quad \text{or} \quad \frac{1}{2} - \frac{1}{4}x > \frac{7}{8}$$

so

$$\left|\frac{1}{2} - \frac{1}{4}x\right| > \frac{7}{8} \iff$$

$$\frac{1}{2} - \frac{1}{4}x < -\frac{7}{8} \quad \vee \quad \frac{1}{2} - \frac{1}{4}x > \frac{7}{8} \iff$$

$$\cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \frac{1}{4}x < \underbrace{\cancel{\frac{1}{2}} - \frac{7}{8}}_{(\times 4)} \quad \vee \quad \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \frac{1}{4}x > \underbrace{\cancel{\frac{1}{2}} + \frac{7}{8}}_{(\times 4)} \iff$$

$$-\frac{1}{4}x < -\frac{4}{8} - \frac{7}{8} \quad \vee \quad -\frac{1}{4}x > -\frac{4}{8} + \frac{7}{8} \iff$$

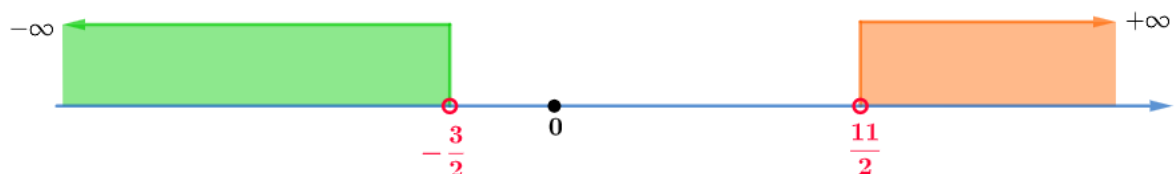
$$-\frac{1}{4}x < -\frac{11}{8} \quad \vee \quad -\frac{1}{4}x > \frac{3}{8} \iff$$

$$\cancel{\frac{1}{4}} \times 4 \times x < -\frac{11}{8} \times 4 \quad \vee \quad \cancel{\frac{1}{4}} \times 4 \times x > \frac{3}{8} \times 4 \iff$$

$$-x < -\frac{11}{2} \quad \vee \quad -x > \frac{3}{2} \iff$$

$$\frac{-x}{-1} > \frac{-\frac{11}{2}}{-1} \quad \vee \quad \frac{-x}{-1} < \frac{\frac{3}{2}}{-1} \iff$$

$$x > \frac{11}{2} \quad \vee \quad x < -\frac{3}{2}$$



Answer: $x < -\frac{3}{2} \wedge x > \frac{11}{2}$ or $x \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{11}{2}, +\infty\right)$

Exercise 6. Sketch the graph of $y = |x| + 3$ and then explain why $|x| + 3 \geq |x + 3|$ for all $x \in \mathbb{R}$.

It helps to write the function $f(x) = |x| + 3$ in the form of a piecewise function.

We know that

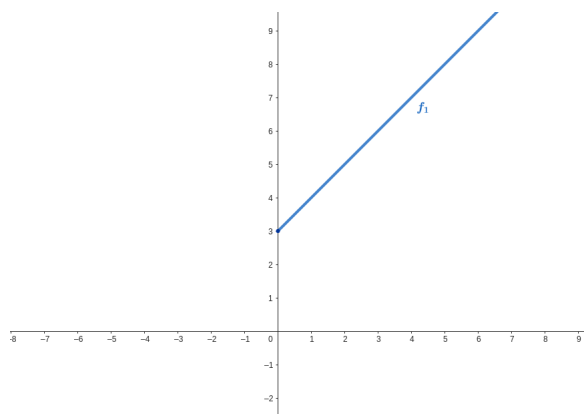
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

so

$$f(x) = |x| + 3 = \begin{cases} x + 3 & \text{if } x \geq 0 \\ -x + 3 & \text{if } x < 0 \end{cases}$$

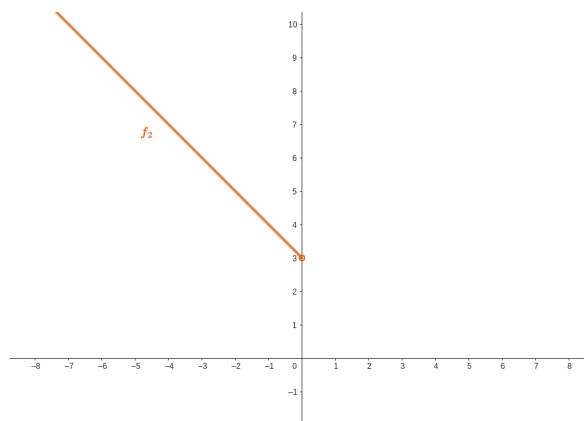
The first sub-function is

$$\begin{aligned} f_1: \mathbb{R}_0^+ &\longrightarrow \mathbb{R} \\ x &\longmapsto x + 3 \end{aligned}$$



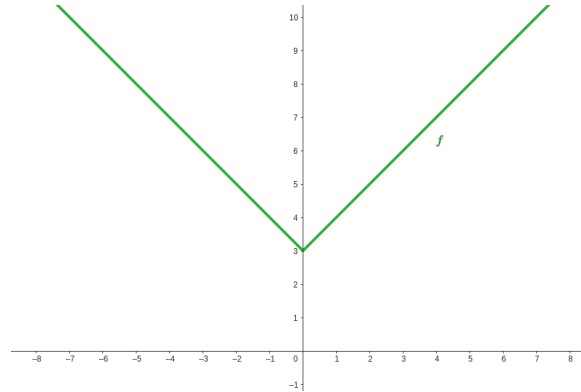
The second sub-function is

$$\begin{aligned} f_2: \mathbb{R}^- &\longrightarrow \mathbb{R} \\ x &\longmapsto -x + 3 \end{aligned}$$



Putting the functions f_1 and f_2 together, we get the graph of the function

$$\begin{aligned} f: \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto |x| + 3 \end{aligned}$$

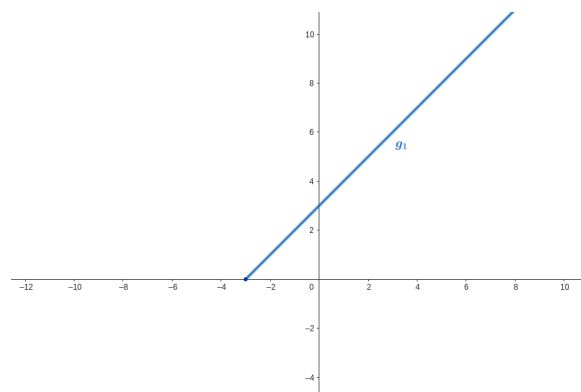


Regarding the function $g(x) = |x + 3|$, we have that

$$\begin{aligned} g(x) = |x + 3| &= \begin{cases} x + 3 & \text{if } x + 3 \geq 0 \\ -(x + 3) & \text{if } x + 3 < 0 \end{cases} = \begin{cases} x + 3 & \text{if } x + \cancel{3} \geq 0 - \cancel{3} \\ -x - 3 & \text{if } x + \cancel{3} < 0 - \cancel{3} \end{cases} = \\ &= \begin{cases} x + 3 & \text{if } x \geq -3 \\ -x - 3 & \text{if } x < -3 \end{cases} \end{aligned}$$

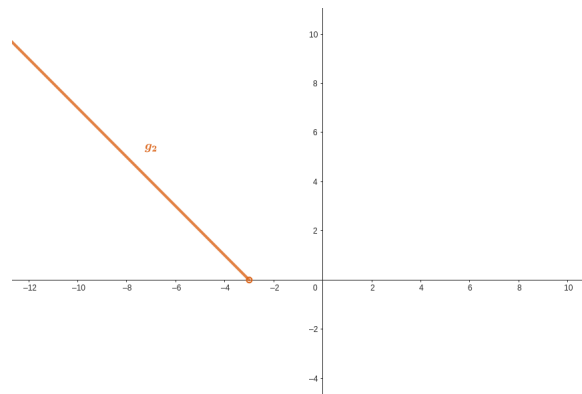
The first sub-function is

$$\begin{aligned} g_1: [-3, +\infty] &\longrightarrow \mathbb{R} \\ x &\longmapsto x + 3 \end{aligned}$$



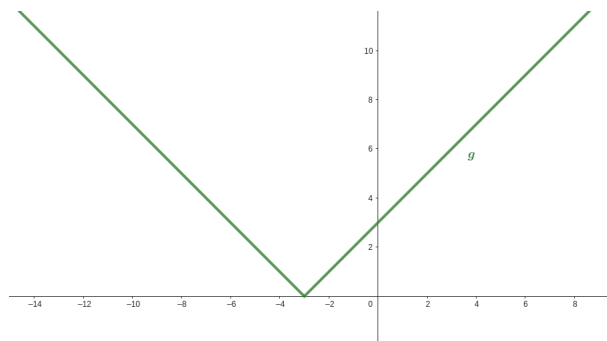
The second sub-function is

$$\begin{aligned} g_2: (-\infty, -3) &\longrightarrow \mathbb{R} \\ x &\longmapsto -x - 3 \end{aligned}$$

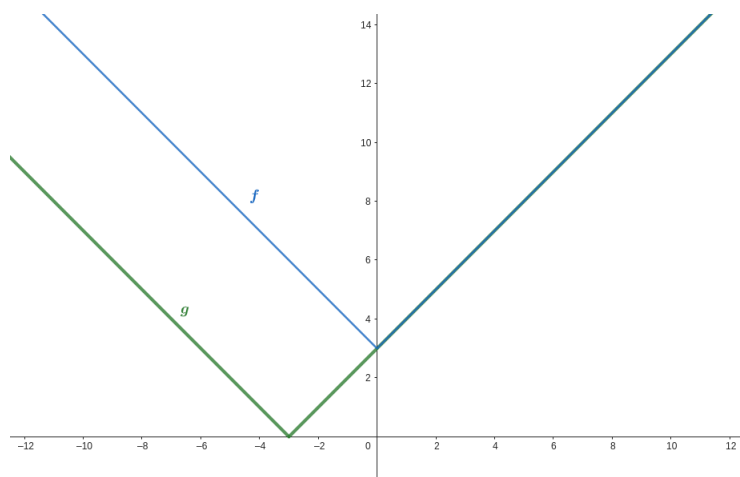


Putting the functions g_1 and g_2 together, we get the graph of the function

$$\begin{aligned} g: \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto |x + 3| \end{aligned}$$



Drawing the graphs of f and g on the same Cartesian plane



it is possible to see that $f(x) \geq g(x)$, $\forall x \in \mathbb{R}$.

Since $f(x) = |x| + 3$ and $g(x) = |x + 3|$, this is equivalent as saying that

$$|x| + 3 \geq |x + 3|, \forall x \in \mathbb{R}$$

Exercise 7. Solve the inequality $|3x - 2| \geq |x + 4|$.

Note the following: Let $a, b \in \mathbb{R}_0^+$ such that $a \geq b$.

Then $a \times a \geq b \times b$, which is equivalent to say that $a^2 \geq b^2$.

Note that here it is essential that both a and b are non-negative real numbers.

Otherwise, the statement can be false. Suppose, for example, that $a = 1$ and $b = -2$.

Then we have $1 \geq -2$, but $1 \times 1 \not\geq -2 \times (-2)$.

Since $|3x - 2| \geq 0$ and $|x + 4| \geq 0$, we have that

$$\begin{aligned} |3x - 2| \geq |x + 4| &\Rightarrow \\ |3x - 2| \times |3x - 2| &\geq |x + 4| \times |x + 4| \iff \\ (|3x - 2|)^2 &\geq (|x + 4|)^2 \end{aligned}$$

In the next step, we can make the modulus disappear because if $3x - 2$ is negative, then when we multiply $3x - 2$ with itself, we get a positive number. So we can say that

$$\begin{aligned} (|3x - 2|)^2 \geq (|x + 4|)^2 &\iff (3x - 2)^2 \geq (x + 4)^2 \stackrel{(1)}{\iff} \\ (3x)^2 + 2 \times (3x) \times (-2) + (-2)^2 &\geq x^2 + 2 \times x \times 4 + 4^2 \iff \\ 9x^2 - 12x + 4 &\geq x^2 + 8x + 16 \iff \\ 9x^2 - x^2 - 12x - 8x + 4 - 16 &\geq \iff 0 \\ 8x^2 - 20x - 12 &\geq 0 \stackrel{(2)}{\iff} \\ 2x^2 - 5x - 3 &\geq 0 \end{aligned}$$

(1) Remember that $(a + b)^2 = a^2 + 2ab + b^2$, $\forall a, b \in \mathbb{R}$.

(2) Divide both sides of the inequality by 4.

Now we have to solve $2x^2 - 5x - 3 \geq 0$.

To do so, we first have to find the roots of the polynomial $2x^2 - 5x - 3$:

$$\begin{aligned} 2x^2 - 5x - 3 &= 0 \iff \\ x &= \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 2 \times (-3)}}{2 \times 2} \iff \end{aligned}$$

$$x = \frac{5 \pm \sqrt{25 + 24}}{4} \iff$$

$$x = \frac{5 \pm \sqrt{49}}{4} \iff$$

$$x = \frac{5 \pm 7}{4} \iff$$

$$x = \frac{5-7}{4} \vee \frac{5+7}{4} \iff$$

$$x = \frac{-2}{4} \vee x = \frac{12}{4} \iff$$

$$x = -\frac{1}{2} \vee x = 3$$

x	$-\infty$	$-\frac{1}{2}$		3	$+\infty$
$2x^2 - 5x - 3$	$+$	0	$-$	0	$+$

As we can see, the solution set of the inequality $2x^2 - 5x - 3 \geq 0$ is

$$\left(-\infty, -\frac{1}{2}\right] \cup [3, +\infty)$$